Content-Addressed Continual Memory with On-Demand Capacity

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Abstract

We study a continual-learning memory in which storage is addressed by content, capacity is allocated on demand from the stream without task identifiers, and new capacity is appended without perturbing anything already stored. On a relational stream whose regimes are distinguishable from their inputs, the mechanism retains 49.5% of probed facts zero-shot against 15.1% for a monolithic sparse-coded control of the same width, which is itself at the level of a linear baseline; the advantage narrows from 34 to 25 points between eight and thirty-two regimes without closing. Sharing the payload transform across regimes while keeping per-regime readouts adds 16.4 points over a fully private variant (six seeds, $p < 0.001$). Allocating readout rows only for targets a regime actually emits reduces occupied capacity by $7.6\times$ at no cost in accuracy, and appending capacity leaves existing weights byte-identical and existing predictions bit-identical.

We give equal space to what the mechanism does not do, and hold those readings open rather than assert them. It carries nothing between facts beyond a single-timescale context, and retention falls as a cue is separated from its query; multi-hop routing occupies the same capacity for less accuracy, and letting the address compose as well as the value costs more still; retention declines as regimes accumulate, with capacity starvation, worsening misrouting and failing separation each excluded by measurement, leaving retrieval difficulty as the surviving account. We report these as measurements whose interpretation is unsettled because, in the course of this study, six measurements that read as findings turned out to have been produced by the measurement rather than the mechanism.

1 Introduction

A learner that must absorb an unbounded stream with bounded parameters has to discard something. Regularisation methods such as elastic weight consolidation [Kirkpatrick et al., 2017] give up plasticity, protecting old weights at the cost of new learning. Methods that simply keep training give up the past. A third option is to *allocate*: give distinct regimes distinct storage, so that degradation is uniform rather than selective. Dynamically expanding architectures [Rusu et al., 2016, Yoon et al., 2018] take this route and are routinely criticised for adding parameters per task without demonstrating that the addition is proportionate to new content rather than to task count.

This paper examines an allocation mechanism built around three commitments. Addressing is by content through a fixed random hash, so a fact reaches the same storage under every history. Regime identity is inferred online from the stream, so no task identifier, task boundary, or replay buffer is required. Capacity is appended rather than resized, so adding storage leaves what is already there untouched, which we check exactly rather than statistically.

We report what these commitments buy and what they cost. They buy a large advantage where regimes are distinguishable, a capacity footprint that follows emitted content rather than vocabulary size, and a per-token cost independent of sequence length. They cost the ability to associate items across a gap, which content-only addressing forbids by construction, and they leave retention declining as regimes accumulate.

Contributions.

1. An allocation mechanism whose two addressing factors — content to a memory slice, inferred regime to a readout — are separately measurable, with no task supervision at any point (§3).
2. Evidence that sharing the payload transform while privatising the readout is the effective split for a stream whose regimes share surface form and differ in targets, confirmed from both directions (§5.2).
3. A capacity result: sampling negatives from a regime's own emitted targets makes occupancy follow content, 7.6× smaller at equal accuracy (§5.3).
4. A non-destructiveness property, verified exactly rather than statistically: appending capacity leaves stored weights and predictions unchanged bit for bit (§5.4).
5. Measurements of what the mechanism does not currently do — cross-gap association, multi-hop addressing, and retention as regimes accumulate — reported with the alternative explanations we could exclude, and left open where we could not (§6).

2 Related work

Continual learning. Catastrophic interference in distributed representations [McCloskey and Cohen, 1989, French, 1999] motivates three broad families [Parisi et al., 2019, De Lange et al., 2021]: regularisation [Kirkpatrick et al., 2017], rehearsal, and architectural expansion [Rusu et al., 2016, Yoon et al., 2018]. Our mechanism is architectural, but differs from prior expansion methods in that it requires no task boundary to decide when to expand; the decision is made from the dispersion of an inferred regime's own statistics. Task-free continual learning [Aljundi et al., 2019] shares this aim.

Conditional computation. Structurally the mechanism is a single-layer, top-1 sparsely-gated mixture of experts [Shazeer et al., 2017, Fedus et al., 2022] whose router is a fixed random hash rather than a learned gate, and whose expert count grows. We treat the unlearned router as a limitation to be reported rather than a feature: it forgoes load balancing and learned specialisation, and it is precisely what prevents context from influencing retrieval (§6).

Addressable and distributed memory. Content-addressed storage with sparse codes follows sparse distributed memory [Kanerva, 1988] and hyperdimensional computing [Kanerva, 2009]. Recoverable association of *pairs*, which a smoothed context summary cannot provide, is available through holographic reduced representations [Plate, 1995] or fast-weight outer products [Ba et al., 2016, Schlag et al., 2021]; we identify this as the missing ingredient rather than claim it.

Consolidation across timescales. Cascades of variables at multiple timescales [Fusi et al., 2005, Benna and Fusi, 2016] target the capacity–lifetime tradeoff of bounded synapses. We tested such a cascade in two roles, as weight consolidation and as input context, and report both outcomes.

Attention and its costs. Self-attention [Vaswani et al., 2017] pays $O(N)$ per token in context access and degrades on material in the middle of long contexts [Liu et al., 2024]. A content-addressed memory pays a cost independent of N; we quantify the crossover in §5.5.

Vocabulary growth. That distinct types grow sublinearly in a Zipf-distributed stream is Heaps' law [Heaps, 1978]; we use it to ask whether allocated capacity tracks content, and report that in our generator it does not, for reasons traced to the generator itself (§6).

3 Method

A fact is a pair (e, r) with target t. Token embeddings are fixed random unit vectors and are never learned. The payload is

$$p_0 \;=\; \mathrm{normalise}\big(\mathrm{emb}[e] + \mathrm{emb}[r]\big). \tag{1}$$

3.1 Two addressing factors

Content to edge. Nodes of a small-world ring hold fixed random keys $k_{n,s}$. The walk selects $s^* = \arg\max_s \langle k_{n,s}, p_0 \rangle$ for H hops. Nothing in this path is learned, so a fact reaches the same edge under every history and routing consistency is 1.000 by construction.

Regime to readout. A slow exponential average c of stream tokens is matched against prototypes, $\gamma = \arg\max_j \langle \pi_j, c \rangle$, selecting a readout slice. No task identifier is used anywhere; the mechanism is never told which regime is live.

3.2 Memory

The edge transform is shared across regimes and applied residually,

$$p \;\leftarrow\; \mathrm{normalise}\big(p + \tanh(E_{\mathrm{edge}}\, p)\big), \tag{2}$$

while the readout R_γ is private per regime and produces logits $R_\gamma\, p_H$. The split follows the generative structure rather than preference: a payload is built from entity and relation, which are common across regimes, whereas targets are exactly what regimes disagree about. Section 5.2 confirms this from both directions.

3.3 Learning

Training is online, single-pass and prequential: predict, take the loss, then write. There is no replay buffer, no task boundary and no second pass. The readout takes a delta-rule step on the target row together with negatives sampled from targets that regime has already emitted; gradients propagate back along the walked path into the shared transform. Routers receive no gradient.

3.4 Allocation

A class is split when the dispersion of its own similarity distribution, $\sigma_\gamma/|\mu_\gamma|$, exceeds a threshold, and the new prototype is placed at the current context. Dispersion is used rather than an outlier test because outlier tests fail in two ways we measured: against a global running distribution they fail once regimes are numerous, since the distribution becomes their mixture; against a class's own distribution they fail because a class absorbing more regimes raises its own variance, so the threshold rises exactly when splitting becomes necessary. Allocation appends, leaving existing slices at their offsets and values.

4 Experimental setup

The source is a multi-domain relational graph walked continuously, with facts stratified by visit frequency into hub, mid and tail tiers under a Zipf distribution. Two modes are used. In **Mode A** each regime owns its entities, so regime identity is present in the input. In **Mode B** entities and relations are byte-identical across regimes and only targets differ, which removes the routing signal and, by construction, any shared target to amortise; we use it as an adversarial bound rather than as the main setting.

Evaluation is zero-shot retention of probed facts under a frozen memory, with the probe restoring the context it needs and returning the state it displaced. All figures in this paper are extracted by a single parser that anchors on the results table header and reads the column by name.

5 Results

5.1 Where regimes are distinguishable

regimes	this work	monolithic	advantage
8	**49.53**%	15.10%	+34.43pp
16	**43.07**%	12.03%	+31.04pp
32	**37.65**%	12.20%	+25.45pp

Table 1: Mode A; three seeds at eight and sixteen regimes, one at thirty-two. The monolithic control sits at the level of a linear baseline without sparse coding (15.30%), so the task is not solvable by predicting frequent targets. The advantage narrows as regimes accumulate but does not close. We did not obtain a monolithic baseline at sixty-four regimes; a single cell costs about nine hours here, and the run was lost.

5.2 Sharing the transform, privatising the readout

Mode B, twelve regimes, six seeds	zero-shot	paired difference
Shared transform, private readouts	**29.42**%	+16.43pp, 6/6, $p < 0.001$
Fully private	12.98%	—
Monolithic	21.28%	+8.13pp, 5/6, $p = 0.119$

Table 2: Sharing the transform is established against the private architecture. The advantage over the monolithic control is not.

Sharing the *readout* as well costs 27.50 points. A common readout block is updated on every write and therefore accumulates every regime's target mapping, while Mode B is defined by the same (e, r) pointing at different targets per regime; it averages contradictions. The asymmetry of §3 is thus confirmed from both sides, one by making the change and the other by testing its opposite.

5.3 Capacity follows content

5.4 Expansion does not disturb what is stored

Appending a class slice leaves every existing weight byte-identical and `forward` bit-identical, asserted as a test rather than measured statistically. A monolithic model cannot offer this: its

	occupied rows	rows per class	accuracy
Dense delta rule	45056/45056 (100%)	4096	32.6%
Sampled from emitted targets	5918/45056 (**13.1**%)	**538**	33.8%

Table 3: A 7.6× reduction in occupied capacity at no cost in accuracy.

capacity is the sparse-coding width, and widening it changes the projection through which every stored code was written.

5.5 Cost per token is independent of context length

Context access costs Hd^2 per token regardless of sequence length, against Nd for attention; the crossover is at $N = d$ and the ratio is 128× at $N = 4096$. Against ourselves: overall activation is 6.178% of memory parameters, of which the edge memory contributes 1024 and the remainder is the readout scoring all V rows. Sparsity currently holds for the expert part only.

6 Open questions

Every negative reading in this study that we examined closely turned out, at least once, to be produced by the measurement rather than by the mechanism (Appendix A). We therefore report the following as measurements whose interpretation is unsettled, rather than as properties of the architecture.

Can the mechanism associate across a gap? Nothing is carried between facts beyond a single-timescale context, so a cue survives $(1-\rho)^{\text{gap}}$.

gap	cue remaining	accuracy
1	0.990	10.10%
16	0.851	10.33%
64	0.526	8.40%
256	0.076	6.83%

Table 4: Paired degradation +3.27pp, 5/6 seeds, $p = 0.013$. The location of the cliff was predicted from the context rate before the run.

Exact association under constant memory is information-theoretically unavailable, so the design question is what a lossy code should retain. A multi-scale average answers *which regime* and cannot answer *which item*. Placing context into the address recovers roughly a tenth of the gap and costs lookup accuracy, consistent with content-only addressing being what forbids association in the first place.

The evaluation itself is unfavourable to the mechanism, and we want that on the record. Each run fixes a single gap, and a single gap is best served by whichever single time constant happens to match it; a multi-scale context can therefore show no advantage under this protocol by construction, because the comparison is decided by the evaluation rather than by the code.

Under a mixed-gap protocol that cycles horizons within a run, the sign of the effect changes: putting the multi-scale trace into the address is worth +0.67pp on average against not doing so, where under a single fixed gap the same setting bought +0.40pp at the long end while costing 0.47pp at the short one. Three seeds and two of three in the favourable direction is far too little to establish anything, but it is consistent with the protocol rather than the code having decided the earlier comparison. We regard both protocols as preliminary: a gap constructed in a synthetic stream fixes what has to be carried and for how long, and cannot show what a real dependency structure demands. Establishing this needs evaluation on natural language at scale, which we have not done.

Whether this is a limit of the mechanism or of this particular probe is not settled. Three earlier versions of the same measurement were invalid, each for a different reason, and the fourth is the one reported here; placing context into the address recovers roughly a tenth of the gap, which suggests the addressing scheme rather than the context code may be the binding constraint.

hops	zero-shot	occupied rows
1	**55.15%**	750
2	46.00%	737
3	29.63%	750

Table 5: Identical occupancy, monotonically worse retention. The under-training explanation is excluded: writes per edge *rise* with hops, from 3246 to 9222, because each fact touches more edges.

Does multi-hop addressing ever pay? We do not conclude that depth is unnecessary, and the setup is narrow enough that we could not. Multi-hop is the only form of depth this architecture has, and it is depth in the transform rather than in the routing: hop $h+1$ transforms hop h's output, but every hop selects its edge with the same fixed copy of p_0, so the sequence of edges a fact visits is settled by its original content. An earlier version of this comparison was taken while regime separation had collapsed; the present one uses a single source, one scale, and a lookup task whose targets are determined entirely by the current fact. A source that rewards composition — where an answer follows from chaining two stored relations rather than retrieving one — is exactly where combinatorial addressing would be expected to pay, and we have not built one. What the table shows is that paths cost accuracy when nothing requires a path.

regimes	zero-shot	occupied	rows/regime	home classes per regime
8	49.53%	530	66	0.75
16	43.07%	1142	71	0.64
32	37.65%	2412	75	0.70
64	34.80%	5198	81	0.78

Table 6: Monolithic baselines are available to thirty-two regimes and are reported in §5.1; the sixty-four-regime cell was not obtained.

Why does retention decline as regimes accumulate? Three explanations are excluded by the data. Capacity is supplied linearly, so this is not starvation. Misrouted write magnitude is 57.0, 54.7, 51.0, 55.0 per cent — flat — so it is not worsening interference. Home classes per regime is flat, so separation does not degrade. What remains is retrieval: confusable stored targets grow 9.8× while accuracy falls to 0.70×, and distinct predictions rise from 47 to 57 rather than collapsing.

These exclusions constrain the explanation without establishing it, and the monolithic baselines beyond sixteen regimes are still running. We state the retrieval account as the surviving candidate rather than as the finding.

Does directed forgetting reclaim anything? We cannot say. Every measurement we made of it is confounded. The earlier ones ran while a per-domain snapshot keyed by ground truth was reaching the training path, and the later one, which added capacity pressure so that a freed slice would be worth something, ran while the allocation criterion was collapsing every regime into a single class — so there were no separate slices to reclaim in the first place. We report no number for it, and note only that a valid test requires contention between regimes for storage *and* a working allocator, which we did not have simultaneously.

Does acquiring a regime become cheaper as regimes accumulate? Later regimes do cost less per unit accuracy in our runs, but our measurement staggers arrival, and arrival order is itself sufficient to produce the effect: a regime arriving late enters a system whose transform and prototypes are already shaped. Separating order from content requires a design we have not run, so we leave the question open rather than answer it from a measurement that cannot distinguish the two (Appendix B).

Is the decline with scale a power law? Retention falls roughly as $N^{-\alpha}$ with $\alpha \approx 0.16$ over the measured range, N being the number of stored items. Whether this exponent is a property of the mechanism, and whether stronger addressing rather than more capacity moves it, is measurable without new machinery.

What should a constant-memory context retain? Exact association across a gap is information-theoretically unavailable under constant memory, so the design question is what to keep. A multi-scale average answers *which regime* and cannot answer *which item.* Sparse retention of salient items, or superposition with approximate unbinding [Plate, 1995, Ba et al., 2016], keeps identity where an average cannot.

Does the plasticity–stability frontier extend or shift? The frontier traced by our mechanism and that traced by the monolithic control barely overlap in plasticity, so at present only reachability can be compared, not position.

7 Discussion

7.1 Why exact context cannot be cheap

To associate any point in a stream with any other exactly, a system must be able to answer for an arbitrary pair, which requires either retaining every state and comparing against them — $O(N)$ memory and $O(N)$ work per query — or precomputing the pairs, $O(N^2)$ in total. Self-attention pays exactly this and buys exactness within its window in return [Vaswani et al., 2017]. The cost is not an implementation artefact; it is what pairwise access costs.

Under constant memory the question changes. Once the stream is longer than what the memory can hold distinctly, some pairs must become unanswerable, and the design decision is which ones. This is the same shape of decision that continual learning already forces on the weights: bounded capacity against an unbounded stream means something is given up, and methods differ in what [Kirkpatrick et al., 2017, Rusu et al., 2016]. Our mechanism extends that choice to the context.

7.2 A fuzzy background instead of a transcript

The position we take, and the one we find natural given the biological case, is that a learner does not need a transcript of its context. A slowly varying state that says *what kind of situation this is* can serve as an anchor for retrieval, leaving the identity of individual past items unrecoverable.

Our context is exactly that: an exponential summary matched against prototypes, which conditions retrieval without recording what occurred.

Our measurements track this division cleanly. The mechanism answers "which regime" well — regimes separate, and retention exceeds the monolithic control by 25 to 34 points — and fails at "which item occurred" as soon as a cue must survive a gap (§6). That is the anchor working as an anchor and failing as a record, which is what it is.

We do not claim this is the only division, or the best one. At least four alternatives are available and differ in what they keep: sparse retention of a few salient items keeps identity and discards time; superposition with approximate unbinding keeps many items at graceful degradation [Plate, 1995]; outer-product fast weights keep pairs at a capacity that scales with d^2 [Ba et al., 2016, Schlag et al., 2021]; and a hybrid in which a fuzzy anchor selects a slice while a small exact buffer covers the recent past would give up long-range exactness only. Which of these is right is an empirical question we have not settled, and the answer may well depend on what the stream is.

7.3 Where this leaves three familiar problems

Catastrophic forgetting. Allocation does not solve interference so much as convert it. Where a shared representation loses old material as new material overwrites it [McCloskey and Cohen, 1989], storage that is separated by regime removes that channel and substitutes another: retrieval among a growing inventory. The substitution appears to be favourable in our measurements — confusable stored targets grow 9.8× while retention falls to 0.70×, and the distinct predictions made rise rather than collapse — but it is a substitution, not a removal, and the cost reappears as scale.

Lost in the middle. Attention degrades on material by its position in the context [Liu et al., 2024]. Content addressing has no position term at all: a fact stored early is reached by the same hash as one stored late, so retrieval carries no positional decay. What decays instead is regime identification, because the context summary is recency-weighted, and what degrades with load is retrieval, because the inventory grows. The degradation is therefore moved off position and onto inventory and recency, which is a different failure surface rather than a smaller one — a prediction that is testable directly, and that we have not tested.

Plasticity and stability. We do not beat the frontier. Our mechanism and the monolithic control barely overlap in plasticity: the control saturates near 0.31 while ours spans 0.32 to 0.92, so at present the honest statement is about reachability, not position. Whether the frontier is extended or merely shifted requires a regime where both operate, which we did not construct.

7.4 Where such a memory might be useful

The properties that survived suggest applications rather than benchmarks.

Absorbing new domains after deployment. Capacity appends without disturbing what is stored, allocation needs no task label, and learning is single-pass, so a deployed model can take on a new domain without a retraining cycle. The exactness of the non-destructiveness matters here more than its size: one can state that adding capacity did not change existing behaviour, rather than measure that it changed little.

Targeted deletion. A regime occupies identifiable slices, so removing one is a local operation rather than a retraining problem. This follows from the same structure as non-destructive expansion and would be the natural test of it; we have not run it, and note only that a monolithic model offers no comparable handle.

Auditable provenance. Every write's destination is determined by a content hash and an inferred regime, both inspectable, so it is possible to say which slice serves which material. A learned router offers no equivalent account of itself.

Long-tail material. Capacity follows emitted content rather than vocabulary size, so rare items occupy their own rows instead of being averaged into frequent ones — which is also why we advise against subword tokenisation below.

We have tested none of these as applications. They are stated as the uses the measured properties would support, not as results.

8 Architectural variants worth trying

Each variant below targets a specific measurement in this paper rather than a general intuition.

Bound scoring to the emitted rows. Inference scores all V rows of the live readout slice, while a class holds 39–41 occupied rows against a vocabulary of 32768. Restricting the argmax to rows the class has emitted, with the remainder contributing a constant, would cut inference cost by nearly three orders of magnitude and would make the activation figure of §5.5 sparse in both halves rather than only in the expert half. Nothing about the mechanism resists this; it is unimplemented, not difficult.

Address hierarchically to bound the candidate set. The decline with scale tracks the number of confusable stored items rather than capacity, misrouting or separation (§6). If that account is right, the fix is not more capacity but a smaller candidate set at retrieval: a coarse regime code selecting a slice, then a within-slice code over that regime's own targets, so that discrimination is against tens of alternatives rather than thousands. This predicts that the exponent α should fall with the depth of the addressing hierarchy, which is directly testable.

Add a term that stores pairs. A smoothed context can report which regime is active and cannot report which item occurred, so no amount of extra timescales will support association across a gap. An outer-product term, $M \leftarrow \lambda M + p_t \otimes p_{t-\Delta}$ retrieved as Mp, stores pairs recoverably at a capacity that scales with d^2 [Ba et al., 2016, Schlag et al., 2021], and is the same algebraic object as the edge transform already present. The open question is whether it can sit alongside content addressing without dissolving the stability that content addressing provides.

Give the router a learned residual. The fixed random hash buys routing consistency of 1.000 and forgoes load balancing and learned specialisation. A learned correction added to the key, small enough that a fact's destination remains stable within a regime, would recover part of what a gated mixture of experts gets from its router [Shazeer et al., 2017] without giving up the property that makes expansion non-destructive.

Let the address compose, not only the value. The value path already composes: hop $h+1$ transforms hop h's output and gradients flow along the walked chain. What repeats is the *address* — every hop selects its edge with the same fixed copy of p_0, so the sequence of edges a fact visits is settled before the walk begins.

We implemented the obvious remedy, copying the transformed payload into the routing key after each hop so that hop $h+1$ selects from hop h's output, and it makes matters worse. Within its own configuration, which uses a larger ring than the multi-hop table above and is therefore not directly comparable to it, accuracy falls from 51.43% to 41.03% at two hops and from 35.10% to 22.83% at three, so the gap to a single hop widens rather than closes. The reason is visible in the same run. Routing consistency, which the fixed key guarantees at 1.000, falls to 0.910; a fact's path now depends on weights that move, so nine per cent of probed facts no longer reach where training left them, and that loss outweighs whatever composition buys. Whether a source that genuinely requires chaining two stored relations would reverse the balance is untested — on a lookup whose target is fixed by the current fact there is nothing for a composed path to find — so this settles the version we tried rather than the idea.

A note on tokenisation. We recommend against subword tokenisation for this architecture. Byte-pair encoding is designed to flatten the frequency distribution, and this mechanism depends on that distribution in two places: allocation splits a class when its similarity dispersion rises, which requires regimes to differ in what they emit, and capacity follows the count of distinct emitted targets, which requires a target to be a unit rather than a sequence. A rare entity split into several frequent subwords loses its identity twice over — its payload becomes a sum of common pieces, which collides in the content hash with every other entity sharing those pieces, and its target becomes a sequence with no single row to occupy. The long tail is exactly where a continual learner has something to gain, and subword vocabularies are built to spend it. A word- or entity-level vocabulary, or any tokenisation that keeps rare units intact, preserves both the Zipf structure the allocator reads and the one-target-one-row correspondence the capacity argument relies on.

9 Limitations

Results are on a synthetic relational source; no natural-language corpus is evaluated, and a stream we constructed fixes what has to be learned and what can be shared, which is precisely what a real corpus would decide for us.

The monolithic control is established at eight, sixteen and thirty-two regimes, with one seed at thirty-two and none at sixty-four. It is not parameter-matched in either direction: at eight regimes it holds 16.8M readout weights against 11.6M allocated by our mechanism, of which about 17K are occupied. The comparison is therefore between two models of comparable width rather than of equal size, and the occupancy figures should be read as describing our mechanism rather than as a like-for-like efficiency claim.

Cost comparisons between the dense and sampled write paths are not commensurable, because write magnitude accumulates over a different number of rows in each. The private and shared arms likewise differ in parameter count, which strengthens the accuracy comparison between them but confounds any comparison of their costs.

Sample sizes are small throughout: six seeds for the sharing result, three for most others, one for the largest control. Several effects in this study weakened or reversed between three seeds and six, so single-digit seed counts should be read accordingly.

Whether acquiring a regime becomes cheaper as regimes accumulate is not settled here: our measurement staggers arrival, and arrival order alone can produce the effect (Appendix B).

10 Conclusion

Content addressing with on-demand allocation buys a large advantage where regimes are distinguishable, a capacity footprint that follows emitted content, and a per-token cost independent of context length, without disturbing what is already stored. The same commitment that makes storage stable — addressing by content alone — is what prevents association across a gap, and the cost of that trade is measured here rather than asserted.

A Measurements invalidated during this study

Six measurements were invalidated after collection and re-run. Three were probe-side leaks in which a per-domain snapshot keyed by ground truth handed the mechanism the regime identity it was meant to infer. One was the allocation criterion of §3.4 in an earlier form, which compared novelty against a global running distribution and collapsed every regime into a single class at scale, voiding the savings, capacity and multi-hop measurements taken under it. One was a parser that scanned past the results table into a table with the same column count and reported 54.2% as 100.00%.

Each was caught by a contradiction between statistics rather than by reading code: a count that disagreed with another count, a diagnostic that printed nothing, an accuracy that could not coexist with its own exposure figures. The instruments that survived — write locality by time since regime change, occupancy against emitted content, allocation timing against regime boundaries — are reported alongside the results because they are what made the failures visible.

B Arrival order and the cost of a late regime

Measuring whether a later regime is cheaper to acquire requires staggered arrival, and staggering introduces a confound we record here for the benefit of anyone running the same measurement.

	shared transform	private
Mode A, staggered arrival	0.754	0.820
Mode B, disjoint targets, staggered arrival	0.442	0.505
Mode A, no staggering (index only)	1.199	1.207

Table 7: Cost of the late regimes relative to the early ones; below one means a late regime is cheaper. Three seeds per control.

Mode B's targets are disjoint across regimes by construction, so nothing there can be amortised, yet the ratio is lower than in Mode A. Removing the staggering removes the effect. A regime arriving late therefore benefits from entering an already-shaped system, independently of whether it shares content with its predecessors, and a ratio below one cannot on its own be read as amortisation.

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